
Mariana Pachón Puentes

Creative technologist and design engineer working across physical computing, interactive systems, digital fabrication, and software. Experienced building prototypes using electronics, sensors, actuators, and custom hardware for installations, experimental products, and participatory experiences.

Located in Barcelona, Spain.

<https://mariana0pachon.github.io/>

Installations

2025. Flora, electronic garden
Hardware, software & sound designer

An **interactive installation** inspired by a natural ecosystem. It is a circular board that features three-dimensional objects that simulate life through **electronic** elements, functioning in unity and cooperation. Movements are powered by **four Arduinos** that communicate with **six sensors** and **nine motors**. Players collaborate to restore the cycle.

Winner of the Amaze Fest 2025 Explorer Award • Exhibited at the 2025 BCN Games Fest • Featured on the Arduino Blog • Winner of the ADCE Student Bronze Award

2025. Wear the Sound
Hardware & software developer

A series of haptic and **sensory interfaces** take the form of reclining supports or **wearables** that can be worn and shared with others, allowing music to be experienced through vibration. Real-time OSC communication synchronized tactile feedback with a live performance.

Exhibited at Mies Van Der Rohe Pavilion in the 2025 Sónar +D • Exhibited at BCN Games Fest 2025

Experiments & Prototypes

2026. Burbúfonos.
A playful installation where two people talk over the phone and receive only bubble signals from their friend.
Built using an **ESP32** board, 2 microphones, and 2 **hacked bubble machines**.

2026. Noise boxes.
A sound/noise experimentation with metal, motors, and **piezoelectric sensors**.

2026. Shrinking theatre.
A prototype of a theatre that shrinks while the main character of the play gets overwhelmed with their thoughts.
Designed and 3D printed 2 **rack & pinion** mechanisms to fit 2 continuous rotation **servo motors**.

2026. Circuit bending.
An investigative experimentation with opening toys and **modifying the electronic connections** between resistances to generate new sound. **Hacking machines** that were not meant to be opened.

2026. Mirrors on wearables.
Design in Rhino, 3D printing, hardware
Collaboration with fashion designer @rubeearth for the 080 Barcelona Fashion Week. Built a mechanism to attach mirrors to motors on clothes and create a windshield visual effect.

Skills

Electronics

Arduino, ESP32, sensors, actuators, motors, OSC wireless communication

Fabrication

3D modeling using Rhino, 3D printing, laser cutting, CNC machine, soldering, rapid prototyping

Programming

React, TypeScript, Python, Svelte, Processing, SQL

Sound design

Ableton Live, Max MSP, PurrData

Awards

2025. Amaze, Explorer Award. Berlin, Germany
Flora, electronic garden.

2024. ADCE Student awards, Bronze winner
Flora, electronic garden.

Residencies

2025. DINACON. Les, Bali, IDN
Memorias desbloqueadas en el trópico
Ephemeral installation that explores the closeness between Colombia (my home) and Indonesia through a map printed on plantain leaf using the BrailleRAP open-source printer. A path traced between Colombia and Bali lights up when people share the experience of eating a tropical fruit.
Built using a LED fluorescent noodle, an **Arduino**, a **capacitive sensor**, map printed with pressure on a plantain leaf.

Exhibitions

2026. 080 Barcelona Fashion Week. Barcelona, Spain
2025. Barcelona Game Fest. Barcelona, Spain
2025. Wear the Sound. Mies van der Rohe Pavilion. Barcelona, Spain
2025. AMAZE. Berlin, Germany
2024. Final thesis presentation. BAU. Barcelona, Spain

Education

2023-2024. Barcelona, Spain
BAU College of Arts & Design
M.A. Audiovisual Innovation & Interactive Environments

2016-2019. Worcester, MA, USA
Worcester Polytechnic Institute
B.S. in Computer Science
High distinction - GPA 3.75/4.0

Professional Experience

Frontend Engineer – TaxGuardian (Freelance) • 2025-2026

Fullstack Engineer – Healthcasts AI / Travoya.ai (Freelance) • 2024-2025

Fullstack Engineer – MojoTech • 2020-2023

JavaScript Instructor – Codeworks • 2023-2024
